

Cheng-Hsiang Chiu

<https://cheng-hsiang-chiu.github.io/>

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QUALIFICATION

- Ph.D. with expertise in parallel and heterogeneous computing and task-graph scheduling
- Programming skills in C++, CUDA, and parallel frameworks with a focus on high-performance system design
- Proven ability to collaborate with cross-disciplinary teams and translate research into efficient solutions

EDUCATION

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|---|---|
| • University of Wisconsin-Madison
<i>Ph.D. in Electrical and Computer Engineering</i> | Madison, Wisconsin, USA
<i>Aug. 2023 – Aug. 2025</i> |
| • École Polytechnique Fédérale de Lausanne
<i>M.S. in Computer Science</i> | Lausanne, Switzerland
<i>Sep. 2013 – Feb. 2016</i> |
| • National Chiao Tung University
<i>M.S. in Communications Engineering</i> | Hsinchu, Taiwan
<i>Sep. 2005 – Aug. 2007</i> |
| • National Chung Cheng University
<i>B.S. in Electrical Engineering</i> | Chiayi, Taiwan
<i>Sep. 2001 – Jun. 2005</i> |

RESEARCH

- **RL Scheduling**: Developed RL scheduling algorithms for work-stealing runtime and CUDA Graph
- **Taskflow**: Developed Pipeflow, AsyncTask, syclflow in Taskflow (<https://taskflow.github.io>).

SELECTED EXPERIENCE

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|---|---|
| • Software Intern at Intel
<i>Developed the implicit SYCL Graph programming model</i> | Austin, Texas, USA
<i>May 2022 - Aug. 2022</i> |
| • Software Intern at Cadence
<i>Accelerated the executions of buffer insertion algorithm by 16%</i> | Austin, Texas, USA
<i>May 2021 - Aug. 2021</i> |

SELECTED PUBLICATIONS

- **C.H. Chiu**, C. Morchdi, C.C. Chang, C. Yu, Y. Zhou, and T.W. Huang, "Optimizing CUDA Graph Scheduling with Reinforcement Learning - A Case Study in SSTA Propagation," *MLCAD*, 2025
- **C.H. Chiu**, C. Morchdi, Y. Zhou, B. Zhang, C. Chang, and T.W. Huang, "Reinforcement Learning-generated Topological Order for Dynamic Task Graph Scheduling", *HPEC*, 2024.
- C. Morchdi, **C.H. Chiu**, Y. Zhou, and T.W. Huang, "A Resource-efficient Task Scheduling System using Reinforcement Learning," *ASP-DAC*, 2024
- **C.H. Chiu**, Z. Xiong, Z. Guo, T.W. Huang, and Y. Lin, "An Efficient Task-parallel Pipeline Programming Framework," *HPC Asia*, 2024
- **C.H. Chiu**, D.L. Lin, and T.W. Huang, "Programming Dynamic Task Parallelism for Heterogeneous EDA Algorithms," *ICCAD*, 2023

SKILLS

- **Programming Model**: Taskflow, SYCL, CUDA, oneTBB, OpenMP, Cilk, Pytorch
- **Programming Language**: C, C++, Python, HTML, SQL
- **Unit Test**: doctest
- **Profiler**: gprof, perf